CASE C

Patient demographics

Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Reduced respiratory muscle endurance
- Generalised deconditioning from infection/inflammation
- Exertional dyspnoea — reduced exercise tolerance
- Fatigue limiting ADL performance
- Reduced participation in community activities

■■ RED FLAGS

- ■ Haemoptysis (frank blood in sputum)
- Indicates possible erosion of bronchial
- artery — a known life-threatening complication of bronchiectasis.
- Volume matters: streaking vs. massive haemoptysis carry different
- Action: Stop ACT immediately. Do NOT perform percussion or vigorous expiratory techniques. Notify medical team immediately.
- ■ SpO■ <92% at rest / Acute severe dyspnoea
- Suggests acute respiratory failure —
- possible from mucus plugging, pneumothorax, or pneumonia superimposed on

■ PATHOPHYSIOLOGY (KEY POINTS)

- Bronchiectasis is characterised by permanent, abnormal dilation and distortion of the bronchi , resulting from a self-perpetuating cycle of infection, inflammation, and airway wall destruction.
- In non-CF bronchiectasis, repeated insults (infection, immune dysfunction) damage the structural components of the bronchial wall — the elastic tissue and smooth muscle — leading to irreversible dilation.
- The dilated airways are structurally abnormal and cannot effectively propel mucus via the mucociliary escalator.
- Mucus stagnates within these pouches and pools, providing an ideal culture medium for bacterial colonisation — commonly Haemophilus influenzae , Pseudomonas aeruginosa , or Staphylococcus aureus .
- Progressive mucus accumulation causes segmental atelectasis and V/Q mismatch , reducing oxygenation and increasing the work of breathing.
- Physiotherapy targets the mucus retention limb of this cycle.

■ DIFFERENTIAL DIAGNOSES

See full case notes

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Sputum Assessment (Colour, Consistency, Volume)

What it measures: Sputum quality and bronchodilator therapy may improve secretion clearance by opening airways. It also gives me a baseline to track improvement, which is important for discharge decisions.

Expected Auscultation:

Widespread coarse crackles — may be more prominent post-spirometry due to increased tidal volume dislodging secretions (a useful productive sign). Possible expiratory wheeze. Diminished air entry at bases if significant atelectasis is present.

Outcome Measure:

FEV■ (% predicted) — repeat at end of admission. Target return toward admission baseline or pre-exacerbation spirometry values if known.

1-Minute Sit-to-Stand Test (1MSTS)

What it measures: Functional exercise capacity and physiological response to exertion — a validated, safe alternative to the 6MWT during acute exacerbation when a corridor walk may be too demanding.

■ TREATMENT PLAN

- completely, as percussion and vigorous ACT are contraindicated.)
- How breathless are you compared to your normal level — are you short of breath getting dressed or walking to the bathroom?
- functional baseline and degree of deterioration from usual status.
- Helps me classify exacerbation severity and decide whether assisted ACT
- or supplemental O■ monitoring is needed.)
- Which airway clearance techniques do you normally use, and what do you feel is stopping them from working as well right now?
- sputum viscosity, fatigue, dyspnoea — so I can target the specific
- problem, whether that is technique modification, positioning, or an
- adjunct device.)
- Do you have any chest pain, particularly when you breathe in deeply or cough?
- chest pain suggests pneumonia or pneumothorax — both change my
- How is the fatigue affecting your ability to do your airway clearance and your usual daily activities?